

CIRCULATORY SYSTEM

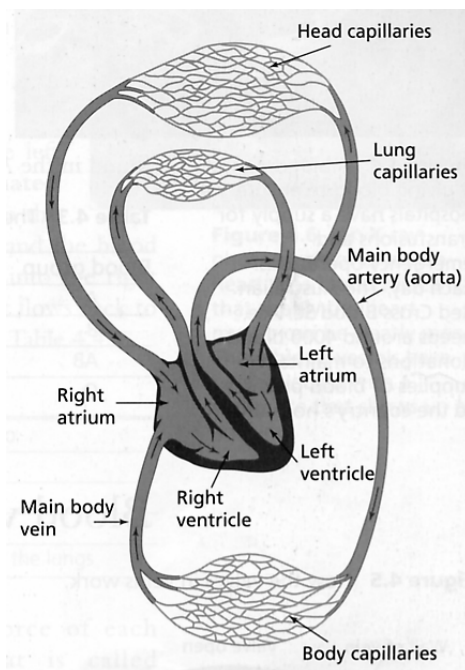
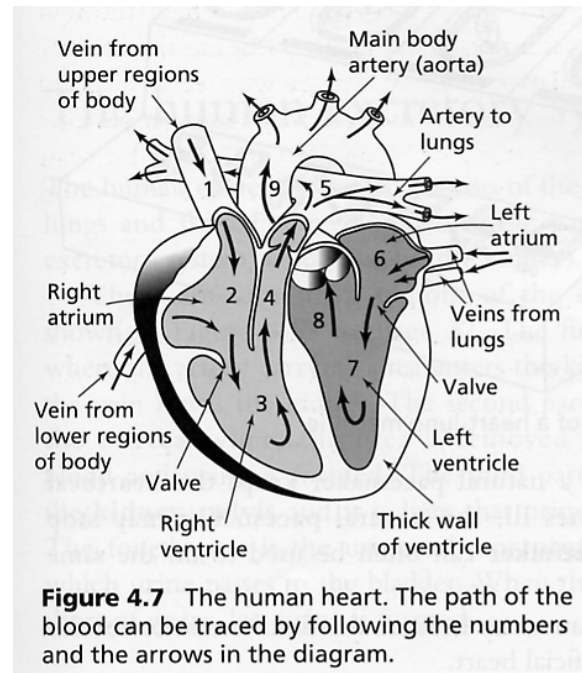
Mammals have a four-chambered heart.

Its structure is as follows.

Atria - collection chambers for blood returning to the heart.

Ventricles - pump blood away from the heart around the body via the arteries.

Valves - keep blood moving in one direction.



Right Side - thin-walled and pumps blood to the lungs to pick up O_2 and give off CO_2 .

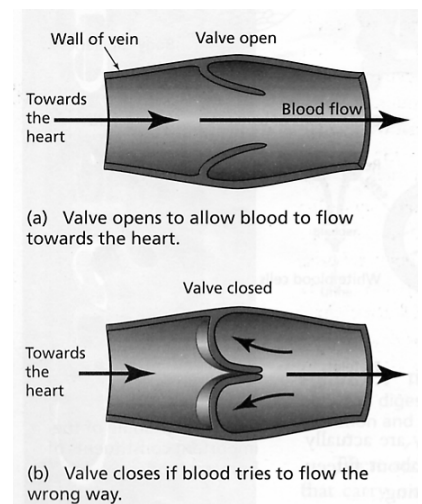
Left Side - thick-walled and pumps blood around the body.

Arteries - thick-walled (since blood pressure is high) and carries blood away from the heart.

Veins - thin-walled (since blood pressure is low) and carries blood back to the heart. Have valves to keep the blood flowing in one direction.

Table 4.1 The main vessels of the circulatory system

Vessel	Structure of vessel wall	Function
Artery	Thick-walled	Carries blood to organs
Capillary	Very thin walls only one cell thick	Allows materials to pass from blood to cells and cells to blood
Vein	Thin-walled	Carries blood back to the heart



Blood

It has the following components.

Plasma - is mainly H_2O and transports O_2 , digested food, CO_2 , wastes, hormones, salts, heat.

Red Cells - carries O_2 on *haemoglobin* it contains.

White Cells - attack bacteria in the blood to protect the body.

Platelets - small cells that help the blood clot in wounds.

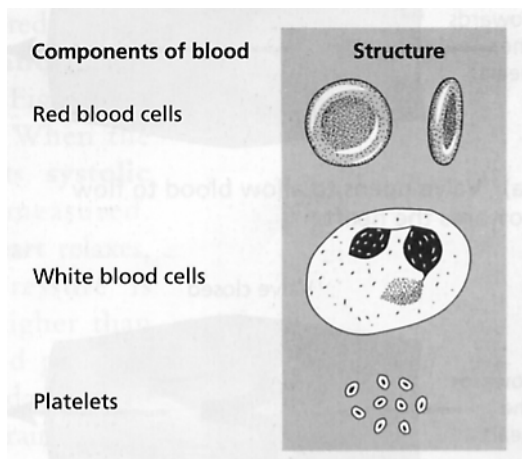


Table 4.2 The main components of blood

Component	Function
Plasma (55%)	Transport of: 1 Oxygen 2 Digested food 3 Carbon dioxide 4 Wastes 5 Hormones 6 Salts 7 Heat
Blood cells (45%)	
1 Red blood cells	Carry oxygen
2 White blood cells	Protect the body
3 Platelets	Clotting of blood